

Yuvraj Sehgal

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github.com/17YuvrajSehgal • linkedin.com/in/yuvraj-sehgal • Portfolio

Computer Science MSc researcher and software engineer with 2+ years of production experience across AI services, full-stack development, data pipelines, and QA automation at Scotiabank and Mengalo. Sole recipient of the Match of Minds research scholarship and author of two peer-reviewed papers (FSE, JSS) on machine learning for systems. Specializes in AI/ML (deep learning, NLP, LLM/RAG), data-intensive backends, and reliable automation with Python, SQL, and cloud infrastructure.

Technical Skills

Languages: Python, Java, SQL, JavaScript/TypeScript, Go, Bash

AI & ML: PyTorch, TensorFlow, scikit-learn, Hugging Face, deep learning, NLP, LLM fine-tuning, prompt engineering, LangChain, LangGraph, RAG

Data & Analytics: ETL/ELT, dimensional modeling, dbt, Databricks, Apache Airflow, Apache Spark, Pandas, NumPy, MapReduce, data warehousing

Databases: PostgreSQL, AlloyDB, MongoDB, Firestore, Neo4j (knowledge graphs)

Cloud & DevOps: GCP (Cloud Run, Compute Engine), Docker, Kubernetes, Terraform, CI/CD (Jenkins, Git, Bitbucket)

Web & Testing: React, Node.js, FastAPI, Spring Boot, REST APIs, Selenium, Appium, RestAssured, Cucumber/BDD, JUnit, Jest, OpenTelemetry

Education

M.Sc. Computer Science | Brock University

Sept 2025 – Sept 2027 (expected)

Thesis-based; research in AI for systems and trustworthy, responsible AI. | St. Catharines, ON

B.Sc. (Honours) Computer Science, Intelligent Systems | Brock University

Jan 2021 – Jan 2025

GPA 87/100 | Dean's Honour List (all years) | Golden Key International Honour Society

Experience

Artificial Intelligence Researcher | Brock University

Feb 2026 – Present

Mitacs Accelerate industrial fellowship with Ciena Corp. | St. Catharines, ON

- **TraceSynth:** developed constraint-guided diffusion models that generate production-quality Linux kernel traces; published at *FSE 2026 (Industry Track)*.
- **LMAT:** contributed to an adaptive tracing approach for efficient system-behavior analysis using language models; published in the *Journal of Systems and Software (2026)*.
- Built an agentic incident-triage system (LangGraph + hypothesis-driven ReAct loop) correlating logs, telemetry, and historical Jira data for automated root-cause diagnosis.
- Curated large-scale microservice trace/log datasets from benchmark systems (DeathStarBench, TrainTicket) on GCP for AIOps research.

AI Developer | Mengalo

Aug 2024 – Dec 2025

Canada

- Built and deployed production AI services and REST APIs (Python, FastAPI) on Docker and GCP for business and customer-engagement automation.
- Designed an LLM- and RAG-powered analytics chatbot (LangChain, Neo4j) for real-time, natural-language querying over operational data.
- Implemented NLP, prompt-engineering, and knowledge-graph extraction pipelines to drive innovation across teams.

Research Assistant | Brock University

Jan 2024 – Jan 2025

St. Catharines, ON

- Built an artifact-documentation tool (Python OOP, OpenAI API, Neo4j) that analyzes conference acceptance probabilities and cut documentation time by ~90%.
- Created an open-source dataset of 2,000+ research artifacts across 40 conferences (projected ~30% research-efficiency gain); sole departmental recipient of the Match of Minds scholarship.

QA Automation Engineer | Scotiabank

May – Aug 2024

Toronto, ON

- Built Android test automation for 15+ scenarios (Java, Selenium, Appium), cutting testing time by over 75% for financial applications.
- Developed API automation libraries (Selenium, RestAssured) and HTML/CSS reporting, cutting validation time ~50%.

Full Stack Developer | Scotiabank

May – Aug 2023

Toronto, ON

- Engineered Python ETL/data pipelines to ingest, transform, validate, and reconcile operational data feeds, cutting manual file handling by 20% and improving data reliability and reproducibility.
- Migrated and transformed up to 350 billion rows of data from Databricks to AlloyDB through large-scale data pipelines, improving query performance by 10x.
- Built the bank's LMS platform with a responsive ReactJS frontend and Node.js/REST backend microservices, deployed to GCP Cloud Run with Firestore persistence.
- Delivered a JUnit/Jest/React Testing Library test suite, raising coverage 15% to meet the bank's 85% mandate.

QA Automation Engineer | Scotiabank

Sept – Dec 2022

Toronto, ON

- Automated 35+ regression scenarios for front-office financial systems (Selenium, Cucumber BDD, Java), reducing manual testing effort ~70%.
- Integrated suites into CI/CD with Jenkins, Git, and Bitbucket for automated deployments and code review.

Publications

Y. Sehgal, S. Patel, M. Panahandeh, N. Ezzati-Jivan, F. Tetreault. TraceSynth: Generating Production-Quality Kernel Traces with Constraint-Guided Diffusion Models. *FSE Companion '26 (Industry Track)*, ACM, pp. 496–506.

doi.org/10.1145/3803437.3805222

K. Darvishi, M. Noferesti, **Y. Sehgal**, N. Ezzati-Jivan. LMAT: An Adaptive Tracing Approach Based on Efficient System Behavior Analysis Using Language Models. *Journal of Systems and Software*, vol. 238, art. 112890, Elsevier, 2026.

doi.org/10.1016/j.jss.2026.112890

Certifications & Additional

Certifications: OAuth2 & OpenID Connect • Java EE: Servlets & JSP • SSL Certificates for Web Developers • Maven Fundamentals • Unit Testing with Jasmine

Awards: Mitacs Accelerate Fellowship (Ciena, \$10K) • Graduate Research Funding (\$29,348) • Match of Minds Scholarship (sole departmental recipient) • Golden Key (top 15%) • Brock Entrance Scholarship (\$6K)

Also: TA — COSC 3P97 Mobile Computing & 3P95 Software Analysis & Testing (2025–26) • Indian citizen w/ Canadian study permit • English, Hindi.